

REMIN MOHAMMED

Robotics and Automation graduate

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Robotics and Automation graduate with experience in embedded systems, ROS 2, PLC, HMI, SCADA, and software development. Skilled in building intelligent automation solutions, robotics systems, and IoT applications. Seeking an entry-level engineering role to develop innovative, scalable technologies.

Work Experience

Intern

2024 - 2024

Mask Automation

- Gained hands-on experience with PLC troubleshooting, HMI, and SCADA systems.
- Assisted in configuration, testing, and maintenance of industrial automation solutions.

Intern

2024 - 2024

IPCS Global

- Worked with PLC configuration and industrial network communication.
- Enhanced troubleshooting and automation system maintenance skills.

Projects

Programmable Macro Keypad (Cheap-Deck)

Built a low-cost programmable macro keypad using microcontroller input mapping.

- Developed custom serial communication between the device and PC application.
- Enabled automation shortcuts for productivity and streaming workflows.

Linux Security Monitoring System (VPS-Monitor)

Developed a Linux VPS intrusion monitoring solution with real-time Discord alerts.

- Monitored SSH logins, file changes, network connections, and sensitive file access.
- Improved server security visibility and incident response.

Eye-Tracking Wheelchair

Designed an assistive mobility system controlled through eye movement detection.

- Implemented gaze tracking using Raspberry Pi camera processing
- Designed an assistive mobility system controlled through eye movement detection.
- Controlled high-power DC motor using BTS7960 drivers

Autonomous Weather Monitoring Robot (TRobot)

Developed an ESP32-based robot for real-time weather monitoring.

- Integrated temperature, humidity, and thunderstorm detection sensors.

Fire Extinguisher Bot

Developed a prototype fire suppression system using targeted acoustic frequencies

- fire suppression robot that uses targeted acoustic frequencies to extinguish small flames without water or chemical agents.

AI Receptionist Assistant Robot (Aimee)

Developed an AI-powered receptionist robot using the Gemini API, enabling natural conversations.

- Integrated Gemini AI to enable real-time conversational interactions and intelligent response generation.
- Implemented emotion-aware communication, enhancing user engagement through human-like speech and expressive responses.

Volunteer Experience

Vice Chair

2024 - 2025

IEEE Student Branch, RIET

- Organized 15+ events and workshops, increasing student participation by 40%
- Collaborated with students, faculty, and industry professionals to improve engagement.
- Contributed to initiatives that increased branch membership and engagement by 30%.

Webmaster

2023 - 2024

IEEE Student Branch, RIET

- Developed and maintained the official IEEE Student Branch website.
- Improved online accessibility and increased user engagement by 50%.
- Managed content updates and provided ongoing technical support.

Core Skills

Languages : Python, C++, Lua, JavaScript, SQL

Robotics: ROS, ROS 2, MoveIt 2, Gazebo, Computer Vision

Embedded Systems: Arduino, ESP32, Raspberry Pi

Industrial Automation: PLC, HMI, SCADA

Tools & Platforms: Git, Linux, MySQL, Blender

Web Technologies: HTML, CSS, JavaScript

Areas: IoT, Sensors & Actuators, System Integration, Industrial Control Systems

Education

IHUB School of Learning

2025 - Present

Postgraduate Diploma Robotic Operating System

Rajdhani Institute of Engineering and Technology

2021 - 2025

Bachelor of Technology in Robotics and Automation Bachelor of Technology in Robotics and Automation

Publications

Automatic Hydroponics Detection System and Robotic Pest Management for Enhanced Crop Cultivation

May 2024

International Journal of Innovative Research in Science, Engineering and Technology (IJIRSET)

Co-authored a research publication on smart hydroponics automation, focusing on sensor-based monitoring and controlled-environment agriculture.